

UART Protocol Project

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PROTOCOL?

Protocol is a specific set of rules used to follow for the establishment of communication between two systems. There are two types of protocols. They are:

- On-Chip
- Peripheral

On-Chip: Communication on chips

Ex :AMBA, AHB, APB, AxI, Wishbone

Peripheral: Peripheral Communication

EX: SPI, UART, I2C

UART PROTOCOL:

UART is a Universal Asynchronous Receiver Transmitter protocol.

→ Why Asynchronous?

In UART, between master and slave there is no shared clock hence it is asynchronous in nature as master shares data at some particular baud rate in serial format and receiver receives the data at a different baud rate in serial format.

→ What is a UART ?

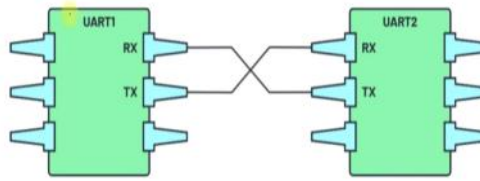
It is a hardware device used for serial communication, commonly found in microcontrollers, computers, and other electronic devices. The master receiving parallel data converts it into serial data and sends it to the receiver which converts the series into parallel to help transmit over communication such as RS-232, RS-485, or TTL, and vice versa.

→ Uses of UART:

- UART is a fundamental component in digital communication systems, enabling serial communication between electronic devices
- Interfacing with Peripheral devices
- Wireless Communication
- Control and Monitoring Systems
- Industrial Automation

WORKING of UART

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- **Parallel to Serial Conversion:**

In UART communication, data is transmitted serially over a communication line(wire) even though it's originally in parallel form on the controlling data bus. The transmitting UART converts parallel data into a serial data stream, sending it bit by bit to the receiving UART.

- **Transmitting and Receiving Pins:**

UART devices typically have dedicated transmit (TX) and receive (RX) pins. The TX pin is used to transmit the data, and the RX pin is used to receive the data. This enables the bi-directional communication between the devices.

- **What is BAUD RATE(or BIT RATE)?**

The Baud rate is expressed as bits per second(bps) is a crucial parameter in serial communication systems like UART.

It determines the rate at which data is transferred between the transmitting and receiving devices over a communication channel.

BAUD RATE: Symbol/Second » bits/second.

It directly influences the data transmission.

Baud Rate is directly proportional to data transmission speed I.e., higher the baud rate = faster data transfer rates. , while lower baud rate results in the speed reduction of data transmission.

*BAUD RATE corresponds to the rate at which bits are transmitted not the number of bits themselves.

In asynchronous serial communication like UART, where there's no shared clock signal between the transmitting and receiving devices, the baud rate serves as a means of synchronization.

Both devices must operate at the same baud rate to ensure that data is transmitted and received correctly.

The transmitting device generates the bitstream based on its clock signal, while the receiving device samples the incoming bits at the same rate using its internal clock.